

Time-Series Crop Masking of Center-Pivot Agricultural Fields in Wadi As-Sirhan Using Sentinel-2 Imagery (2020-2024)

Project Report

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1.Introduction

Center-pivot irrigation creates circular agricultural fields that can be clearly recognised in satellite imagery. In arid regions, these irrigated fields contrast strongly with the surrounding bare desert surface and show substantial seasonal changes in vegetation condition. This makes them suitable for monitoring active agricultural land using multi-temporal optical satellite data and vegetation indices.

This study investigates the seasonal variation of active center-pivot agriculture in the Wadi As-Sirhan region of Saudi Arabia between January 2020 and December 2024. A reproducible Google Earth Engine workflow was developed using Sentinel-2 Level-2A surface reflectance imagery. The workflow applied Scene Classification Layer (SCL)-based quality masking, calculated NDVI for every valid Sentinel-2 observation, generated one monthly median NDVI composite for each calendar month, and converted the composites into binary crop masks using a fixed NDVI threshold.

The methodological approach used in this study is referred to as an SCL-masked, rule-based NDVI thresholding workflow based on monthly median composites. It is a binary classification approach in which pixels with NDVI values equal to or greater than 0.40 are classified as active crop or vegetation pixels, while pixels below this threshold are classified as non-crop pixels. Since no training data were used to train a statistical or machine-learning model, the method is not a supervised classification; instead, it is a transparent and reproducible rule-based approach.

1.1 Objectives

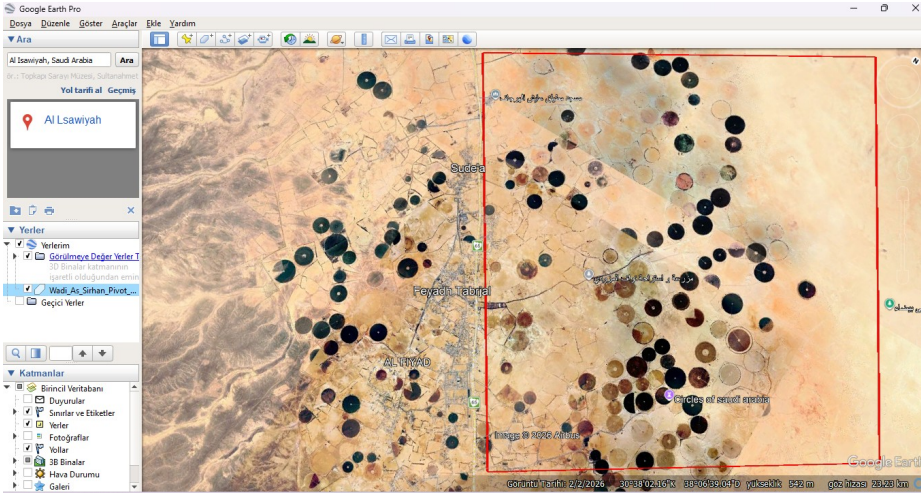
The main objective of this study was to develop and evaluate a five-year Sentinel-2 time-series crop-masking workflow for center-pivot agricultural fields in Wadi As-Sirhan. More specifically, the study aimed to: query and evaluate Sentinel-2 Level-2A image availability between 2020 and 2024; apply SCL-based cloud, cirrus, and shadow masking; calculate NDVI and EVI from the appropriate Sentinel-2 spectral bands; create 60 monthly median NDVI composites; generate binary crop masks using a fixed NDVI threshold; quantify monthly cropped area in square kilometres; compare crop-mask results for off-season, transition-season, and peak-season conditions; and perform a light quantitative validation using manually labelled reference polygons.

2. Study Area

The study area is located in the Wadi As-Sirhan agricultural region of Saudi Arabia, near the settlements of Sude'a and Feyadh Tabarjal. This region was selected because it contains a high concentration of center-pivot agricultural fields, which are clearly recognisable by their circular geometry in satellite imagery.

The Area of Interest (AOI) was delineated as a polygon in Google Earth Pro. The selected AOI covers approximately 190 km² and contains numerous active, inactive, and partially vegetated pivot fields. Most settlement areas and the main road corridor were excluded from the polygon in order to focus the analysis on agricultural land.

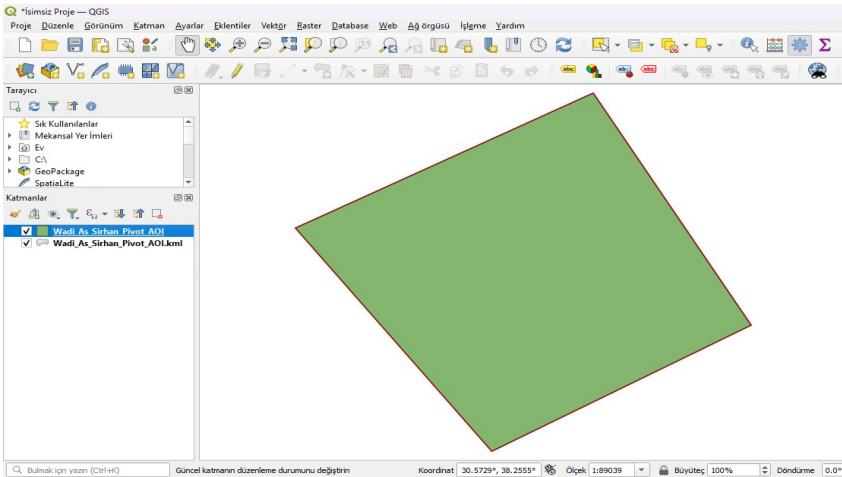
Figure 1. Google Earth Pro view of the selected Area of Interest in Wadi As-Sirhan. The red polygon shows the boundary used for the crop-masking analysis.



This area is especially suitable for time-series crop masking because irrigated fields show strong seasonal changes in vegetation condition. In addition, the contrast between irrigated agricultural fields and the surrounding dry desert surface is expected to improve the separability of active crops using vegetation indices such as NDVI.

The original AOI boundary was saved in KML format and then converted to GeoJSON format in QGIS using the WGS 84 coordinate reference system (EPSG:4326). The GeoJSON polygon was prepared for use in the Google Earth Engine workflow.

Figure 2. Final AOI polygon prepared in QGIS and exported as a GeoJSON file in WGS 84 (EPSG:4326).



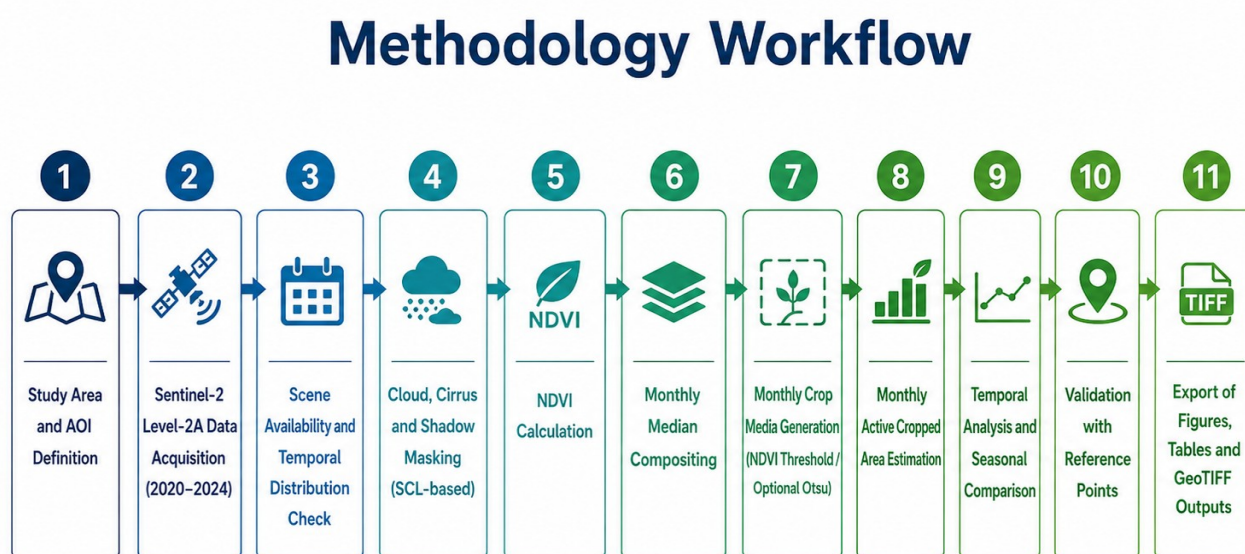
3. Methodology

The workflow was implemented in Google Earth Engine using the COPERNICUS/S2_SR_HARMONIZED Sentinel-2 Level-2A surface reflectance collection. The analysis used 10 m spatial resolution data and covered the period from January 2020 to December 2024. The main method was an SCL-masked, rule-based NDVI thresholding approach based on monthly median composites. NDVI was used as the main crop-masking index, while EVI was calculated as an additional supporting vegetation index. A fixed threshold of $NDVI \geq 0.40$ was applied consistently to generate binary monthly crop masks across the five-year period.

Table 1. Key processing parameters used in the crop-masking workflow.

Parameter	Selected value
Processing platform	Google Earth Engine
Sentinel-2 collection	COPERNICUS/S2_SR_HARMONIZED
Period	January 2020–December 2024
Spatial resolution	10 m
Valid SCL classes	4, 5 and 6
Main vegetation index	NDVI
Supporting index	EVI
Monthly compositing	Per-pixel median
Crop-mask rule	$\text{NDVI} \geq 0.40$
Validation	April 2023, 200 manually labelled pixels

Figure 3. Overall methodology workflow for generating monthly active crop masks from multi-year Sentinel-2 imagery.



3.1 Area of Interest Definition

The Area of Interest (AOI) was defined as a polygon over center-pivot agricultural fields in Wadi As-Sirhan, Saudi Arabia. The boundary was created in Google Earth Pro, converted to GeoJSON in QGIS, and uploaded to Google Earth Engine. The AOI covers approximately **190.06 km²** and includes numerous circular pivot fields while excluding most settlements and the main road corridor.

Figure 4. AOI displayed in Google Earth Engine. The red polygon represents the selected center-pivot agricultural area in Wadi As-Sirhan, Saudi Arabia.



3.2 Sentinel-2 Data Access and Scene Availability

Google Earth Engine was selected because it provides direct access to the Sentinel-2 Level-2A archive and enables cloud-based processing of the five-year image collection.

All Sentinel-2 Level-2A surface reflectance scenes intersecting the AOI were queried in Google Earth Engine for the period from January 2020 to December 2024. No cloud-based filtering was applied at this stage because cloud and shadow masking was handled separately in the following processing step.

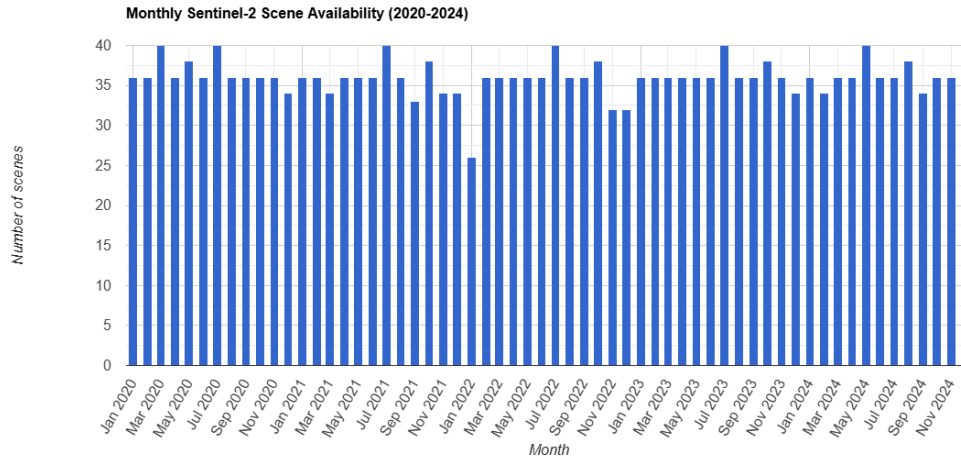
A total of **2,158 Sentinel-2 scenes** were identified over the five-year period. Annual scene availability ranged from approximately **418 to 440 scenes**, while monthly availability ranged from **26 to 40 scenes**. No month lacked Sentinel-2 observations.

The results confirm continuous and sufficient image availability for generating monthly NDVI composites and monthly crop masks throughout the study period.

Figure 5. Annual number of Sentinel-2 Level-2A scenes available over the study area between 2020 and 2024.



Figure 6. Monthly availability of Sentinel-2 Level-2A scenes over the study area between January 2020 and December 2024.

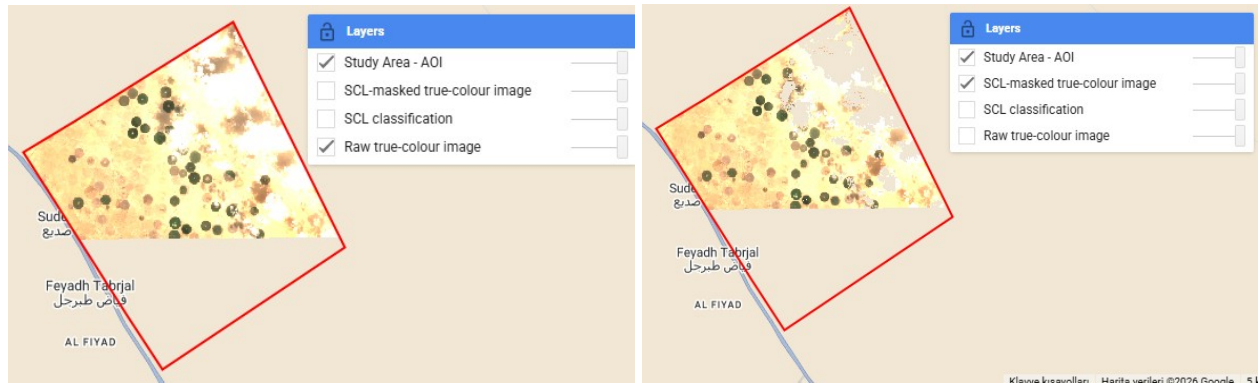


3.3 Cloud, Cirrus and Shadow Masking

An SCL-based masking procedure was applied to every Sentinel-2 Level-2A scene before further analysis. Instead of using a cloud-probability threshold, only pixels classified as vegetation (SCL = 4), bare soil (SCL = 5), and water (SCL = 6) were retained. Cloud shadows, cloud classes, cirrus, snow/ice, and uncertain pixels were excluded.

This conservative class-selection rule was used to prevent cloud-contaminated pixels from affecting NDVI values and causing false crop-mask results. Figure 7 compares a raw true-colour image with its SCL-masked version.

Figure 7. SCL-based cloud, cirrus, and shadow masking for a representative Sentinel-2 Level-2A image: (a) raw true-colour image and (b) SCL-masked true-colour image.



3.4 Vegetation Index Calculation

3.4.1 NDVI Calculation

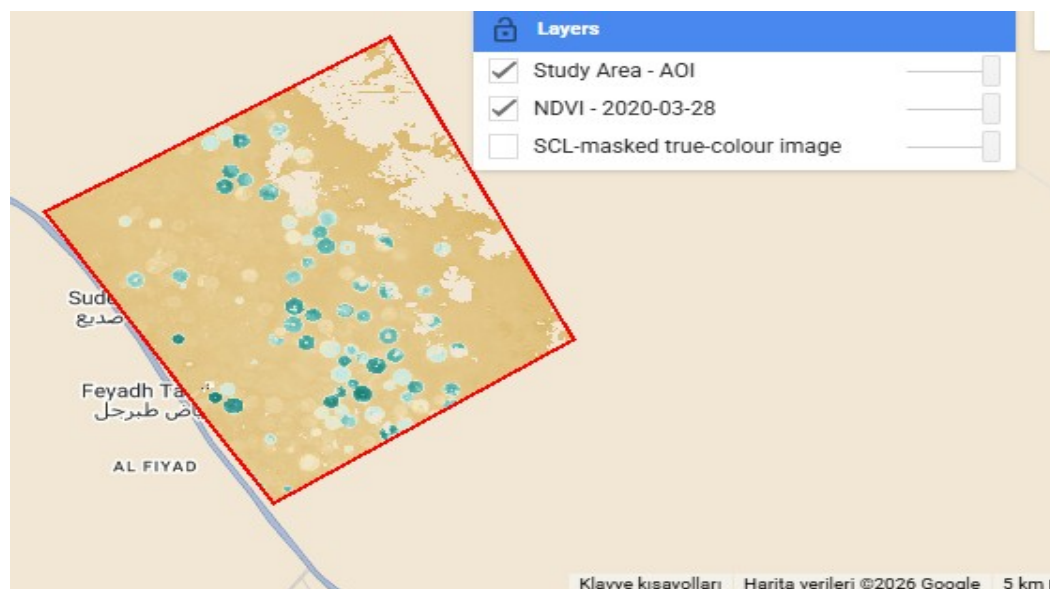
The Normalized Difference Vegetation Index (NDVI) was calculated for each SCL-masked Sentinel-2 Level-2A observation using the Near Infrared (B8) and Red (B4) bands. NDVI was used to distinguish actively vegetated agricultural fields from bare soil and desert surfaces.

$$\text{NDVI} = (\text{B8} - \text{B4}) / (\text{B8} + \text{B4})$$

where B8 is the Near Infrared band and B4 is the Red band of Sentinel-2.

Figure 8 shows a representative NDVI image acquired on 28 March 2020. Circular center-pivot fields are visible in turquoise to dark green tones, indicating relatively high vegetation activity. In contrast, surrounding desert and bare-soil areas show low NDVI values in light yellow and brown tones. This result demonstrates that NDVI can effectively separate active crop areas from non-vegetated land and provides the basis for the monthly crop-mask generation step.

Figure 8. NDVI map derived from a SCL-masked Sentinel-2 Level-2A image acquired on 28 March 2020. Turquoise and dark green tones indicate higher vegetation activity, while light yellow and brown tones represent low-NDVI bare soil or desert surfaces.



3.4.2 Additional Vegetation Index: EVI

As an additional vegetation index, the Enhanced Vegetation Index (EVI) was calculated for all SCL-masked Sentinel-2 scenes using the NIR (B8), Red (B4), and Blue (B2) bands. EVI was included as a supporting indicator of vegetation activity in addition to NDVI.

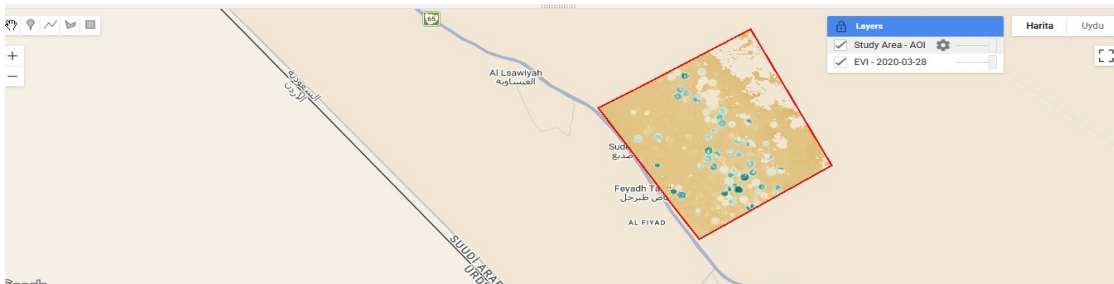
$$\text{EVI} = 2.5 \times (\text{B8} - \text{B4}) / (\text{B8} + 6\text{B4} - 7.5\text{B2} + 1)$$

where B8, B4, and B2 represent the Near Infrared, Red, and Blue Sentinel-2 bands, respectively.

Figure 9 shows the EVI result for 28 March 2020. The circular center-pivot fields appear mainly in turquoise and dark green tones, indicating relatively high vegetation activity. In contrast, the surrounding desert and bare-soil areas have low EVI values and appear in light yellow and brown tones. This confirms that vegetated agricultural fields can be clearly distinguished from the surrounding non-crop land.

EVI was used as a supporting index only; NDVI remained the main input for monthly crop-mask generation.

Figure 9. EVI map derived from a SCL-masked Sentinel-2 Level-2A image acquired on 28 March 2020. Turquoise and dark green tones represent higher vegetation activity, while light yellow and brown tones indicate low vegetation activity or bare desert surfaces.



3.5 Monthly NDVI Compositing

3.5.1 Monthly Data Availability and Composite Preparation

Before generating the monthly NDVI composites, Sentinel-2 scene availability was evaluated for every calendar month between January 2020 and December 2024. This period contains 60 months in total.

The results showed that all 60 months contained Sentinel-2 observations. No month was identified as a no-data month. Monthly scene availability remained consistently high throughout the study period, generally ranging between approximately 32 and 40 scenes per month. For example, March 2020 contained 40 Sentinel-2 scenes.

This result confirms that the Sentinel-2 archive provides sufficient and continuous temporal coverage for producing monthly NDVI composites across the full five-year period. The monthly scene-availability analysis was also used to identify potential data gaps before crop-mask generation; the full monthly distribution is presented in **Figure 6**.

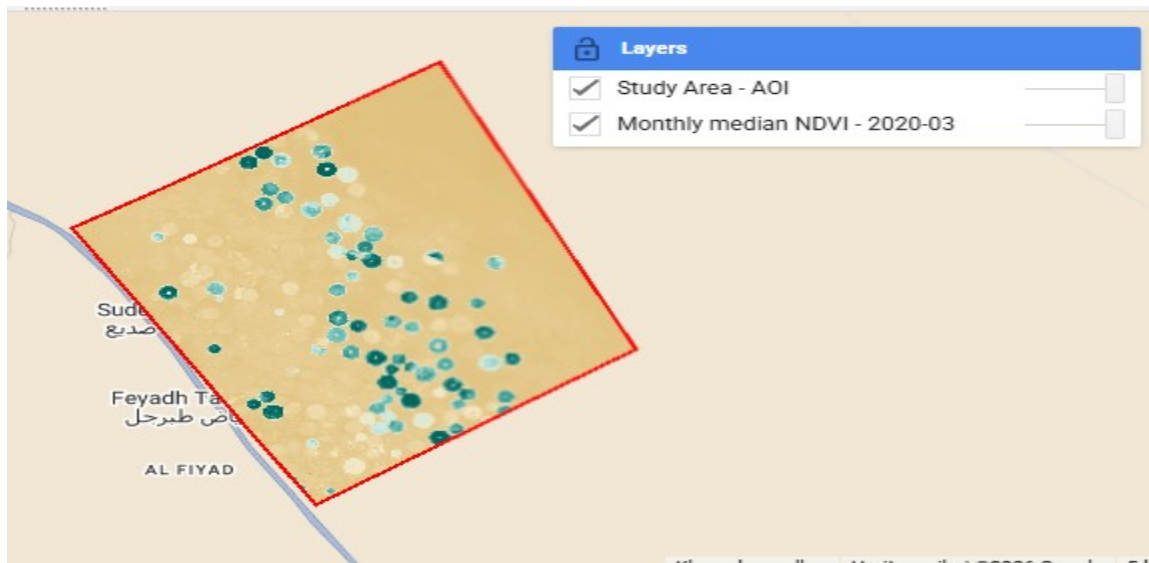
3.5.2 Monthly Median NDVI Composite

After confirming continuous Sentinel-2 data availability, monthly NDVI composites were generated. For each calendar month, all SCL-masked NDVI observations were combined using the median value of each pixel.

Figure 10 presents the monthly median NDVI composite for March 2020. This composite was created from 40 Sentinel-2 scenes acquired during the month. The use of the median reduces the influence of remaining cloud contamination, shadows, and extreme pixel values, producing a more representative NDVI image for the month.

The circular center-pivot fields remain visible as turquoise and dark green areas, indicating higher vegetation activity. In contrast, the surrounding desert and bare-soil surfaces show lower NDVI values. This monthly composite provides the input data for the monthly crop-mask generation step.

Figure 10. Monthly median NDVI composite for March 2020, produced from 40 SCL-masked Sentinel-2 observations. Higher NDVI values over circular pivot fields indicate active vegetation, whereas lower values represent surrounding desert or bare-soil surfaces



3.6 NDVI Threshold Selection for Monthly Crop-Mask Generation

A fixed NDVI threshold approach was tested to generate a binary monthly crop mask. Pixels with NDVI values equal to or above the selected threshold were classified as crop or active vegetation (value = 1), whereas pixels below the threshold were classified as non-crop land (value = 0).

To select a defensible threshold, three candidate values were evaluated using the March 2020 monthly median NDVI composite: 0.35, 0.40, and 0.45. Each candidate mask was visually compared with the monthly true-colour composite and with the visible circular center-pivot fields.

The threshold of 0.35 produced the largest mapped crop area, 16.28 km². It captured a broad extent of the pivot fields but also classified more low-vegetation pixels as crop. The threshold of 0.45 produced the smallest crop area, 14.20 km². Although it selected the most strongly vegetated parts of the fields, it omitted portions of several pivot fields. The intermediate threshold of 0.40 produced a crop area of 15.22 km² and showed the most balanced visual agreement with the center-pivot agricultural fields.

Therefore, $NDVI \geq 0.40$ was selected as the fixed threshold for the generation of monthly binary crop masks for the 2020–2024 period. This threshold was retained as a single documented parameter to ensure that the method remains reproducible and comparable across all months.

Table 2. Comparison of candidate NDVI thresholds for March 2020.

NDVI threshold	Crop area in March 2020 (km ²)	Visual interpretation
≥ 0.35	16.28	Broad selection; includes more low-vegetation pixels
≥ 0.40	15.22	Balanced representation of pivot fields
≥ 0.45	14.20	More restrictive; omits parts of some pivot fields

Figure 11. Comparison of crop masks generated from the March 2020 monthly median NDVI composite using three candidate thresholds: (a) $\text{NDVI} \geq 0.35$, (b) $\text{NDVI} \geq 0.40$, and (c) $\text{NDVI} \geq 0.45$. The $\text{NDVI} \geq 0.40$ threshold was selected for the five-year monthly crop-mask workflow because it provided the most balanced visual representation of center-pivot agricultural fields.



3.7 Crop-Area Time-Series Analysis

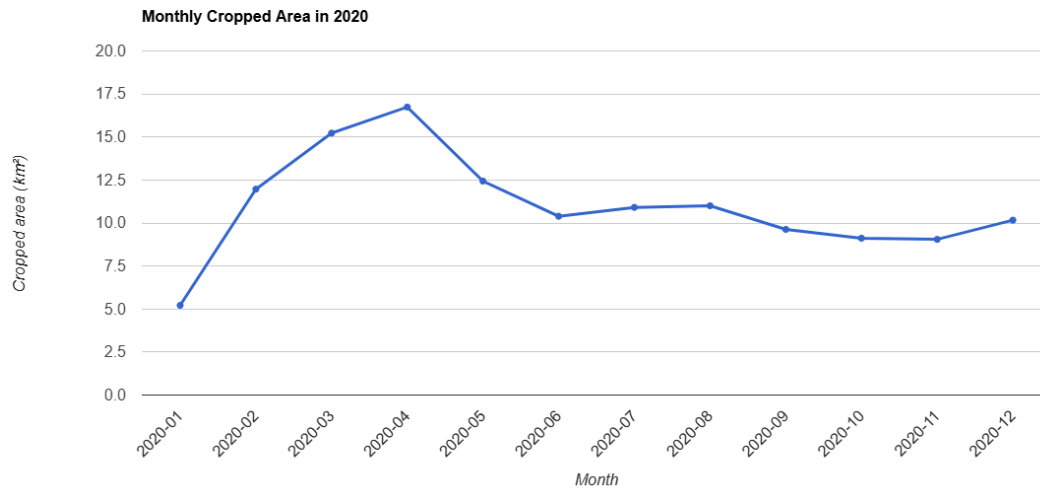
3.7.1 Pilot-Year Test: Monthly Crop Area in 2020

Before scaling the workflow to the complete five-year period, the monthly crop-mask procedure was tested for the 2020 calendar year. For each month, a monthly median NDVI composite was generated from SCL-masked Sentinel-2 observations. Pixels with NDVI values equal to or greater than 0.40 were classified as crop, and the total area of crop pixels was calculated in square kilometres.

Figure 12 shows the resulting monthly cropped-area time series for 2020. The mapped crop area increased from approximately 5 km² in January to a peak of about 17 km² in April. It then decreased during the following months and remained at approximately 9–10 km² toward the end of the year.

This pilot result demonstrates that the fixed NDVI threshold produces a measurable seasonal crop-area signal over the center-pivot fields. It also confirms that the processing workflow can be applied consistently to the remaining years of the study period.

Figure 12. Monthly cropped area derived from binary NDVI-based crop masks for the 2020 pilot year. Crop pixels were defined using $\text{NDVI} \geq 0.40$.



3.7.2 Full Five-Year Crop-Area Time Series (2020–2024)

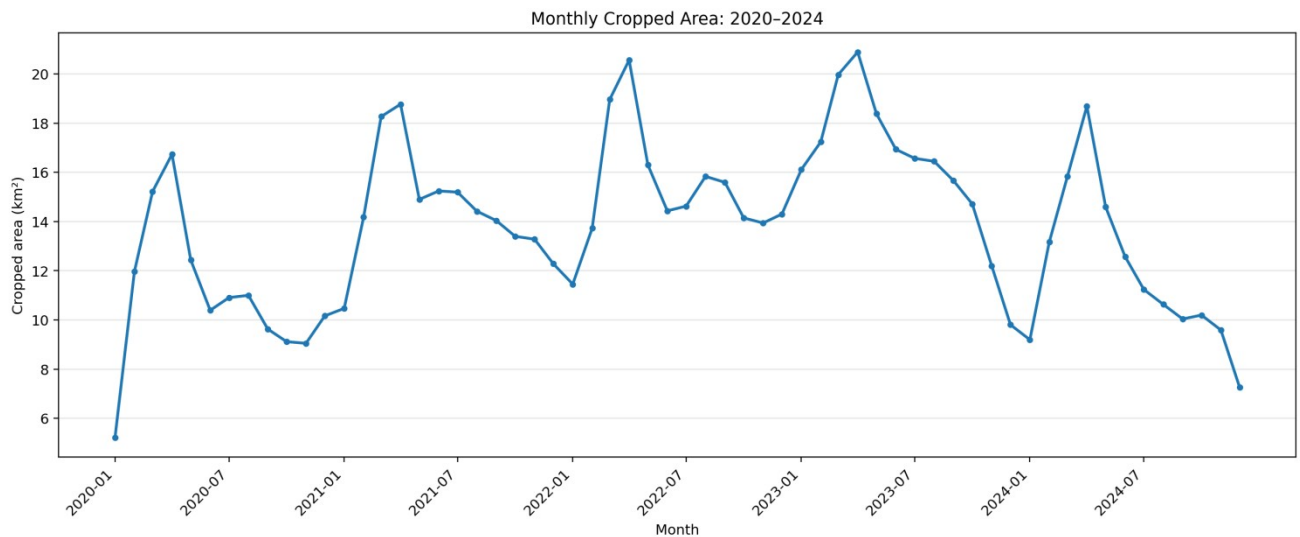
Following the 2020 pilot-year test, the same monthly crop-mask workflow was applied to the remaining years from 2021 to 2024. For each year, Sentinel-2 Level-2A observations were SCL-masked, monthly median NDVI composites were generated, and pixels with NDVI values equal to or greater than 0.40 were classified as crop or active vegetation.

The five-year analysis was processed in yearly batches to keep the workflow computationally manageable and to preserve intermediate results. Each yearly batch produced a 12-month crop-area table. The five annual tables were then merged into one final dataset containing 60 monthly crop-area records from January 2020 to December 2024. The same AOI, SCL masking rule, NDVI calculation, spatial resolution, and NDVI threshold were used for every year to ensure consistency.

Figure 13 shows the final monthly cropped-area time series for the full study period. The mapped crop area ranged from approximately 5.21 km² in January 2020 to 20.90 km² in April 2023. A recurring seasonal pattern is visible across the five years: cropped area generally increases from the beginning of the year toward spring, reaches relatively high values around March or April, and decreases during the later months.

These results indicate that vegetation activity over the center-pivot fields is not constant throughout the year. Instead, the NDVI-based crop masks capture a clear seasonal variation in active agricultural land. The full five-year time series was also used to select representative peak-season, off-season, and transition-season months for the subsequent visual comparison and validation step.

Figure 13. Monthly cropped area derived from NDVI-based binary crop masks between January 2020 and December 2024. The same fixed threshold of $\text{NDVI} \geq 0.40$ was applied consistently to all months.



3.8 Visual Comparison of Representative Seasonal Crop Masks

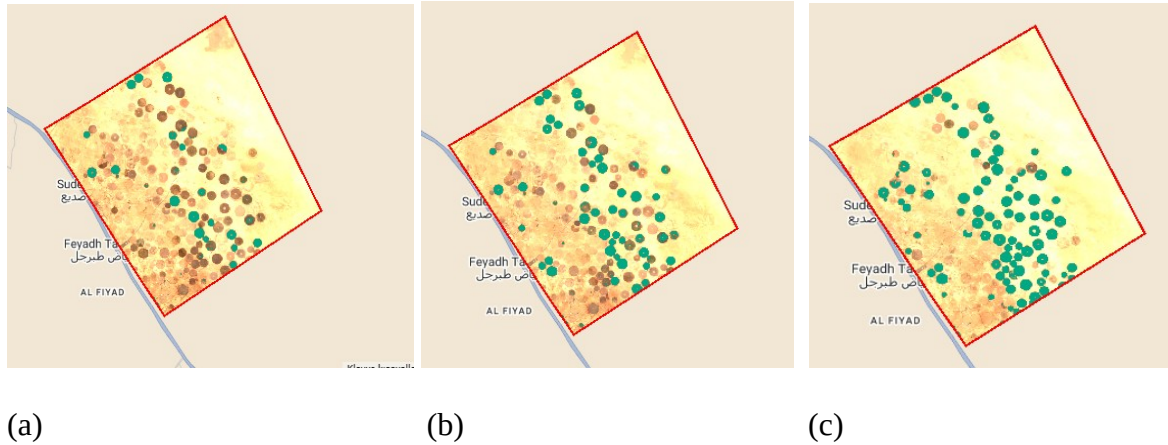
To complement the five-year temporal analysis presented in Section 3.7.2, three representative months were selected from the final monthly crop-area time series. The selected months represent three different seasonal conditions: low activity, transition, and peak activity. January 2020 was selected as the off-season example because it had the lowest mapped crop area in the full time series, with approximately 5.21 km². February 2020 was selected as the transition-period example because the mapped crop area increased substantially to approximately 11.97 km². April 2023 was selected as the peak-season example because it recorded the highest mapped crop area, approximately 20.90 km².

For each selected month, an SCL-masked monthly median true-colour composite was generated using the Red, Green, and Blue bands (B4, B3, and B2). The corresponding NDVI-based binary crop mask was then overlaid on the true-colour composite. Pixels with NDVI values equal to or greater than 0.40 are shown as active crop or vegetation pixels.

Figure 14 provides a visual comparison of the crop-mask results under different seasonal conditions. In January 2020, only a limited number of center-pivot fields were classified as active crop areas, which is consistent with the low crop-area value observed in the time-series graph. In February 2020, the number of detected active fields increased, indicating the beginning of seasonal vegetation development. In April 2023, the crop mask covered the largest number of center-pivot fields, corresponding to the maximum crop-area value in the five-year analysis.

The spatial distribution of the detected pixels is visually consistent with the circular center-pivot agricultural fields visible in the true-colour composites. Therefore, this comparison provides a qualitative visual check that the NDVI thresholding approach captures meaningful seasonal changes in agricultural activity. The representative binary crop masks and true-colour composites were also exported as GeoTIFF files for the final project deliverables.

Figure 14. Visual comparison of SCL-masked monthly true-colour composites and NDVI-based crop-mask overlays for (a) off-season January 2020, (b) transition-season February 2020, and (c) peak-season April 2023. Turquoise pixels represent areas classified as active crop or vegetation using the fixed threshold $NDVI \geq 0.40$.



3.9 Light Quantitative Validation and Limitations

3.9.1 Validation Method and Reference Polygon Selection

A light quantitative validation was carried out to assess whether the NDVI-based crop mask correctly distinguishes active center-pivot fields from non-crop desert surfaces. This validation was performed for April 2023, which was selected as the reference month because it represented the peak-season condition and recorded the highest mapped crop area in the five-year time series, approximately 20.90 km².

Manual reference polygons were created using the April 2023 true-colour Sentinel-2 composite. The crop-mask overlay was not used during polygon selection in order to avoid influencing the reference labels. Two small polygons were placed inside homogeneous and clearly active center-pivot fields and labelled as crop reference areas. Two additional polygons were placed over homogeneous bare desert surfaces, away from pivot fields, roads, and transition zones, and labelled as non-crop reference areas.

The manually selected polygons were then used as reference data for a light quantitative accuracy assessment. A maximum of 50 pixels was randomly sampled from each reference polygon. Therefore, the validation dataset contained 100 crop reference pixels and 100 non-crop reference pixels, resulting in a total of 200 validation pixels. The manually assigned reference class of each

sampled pixel was compared with the predicted class produced by the NDVI-based binary crop

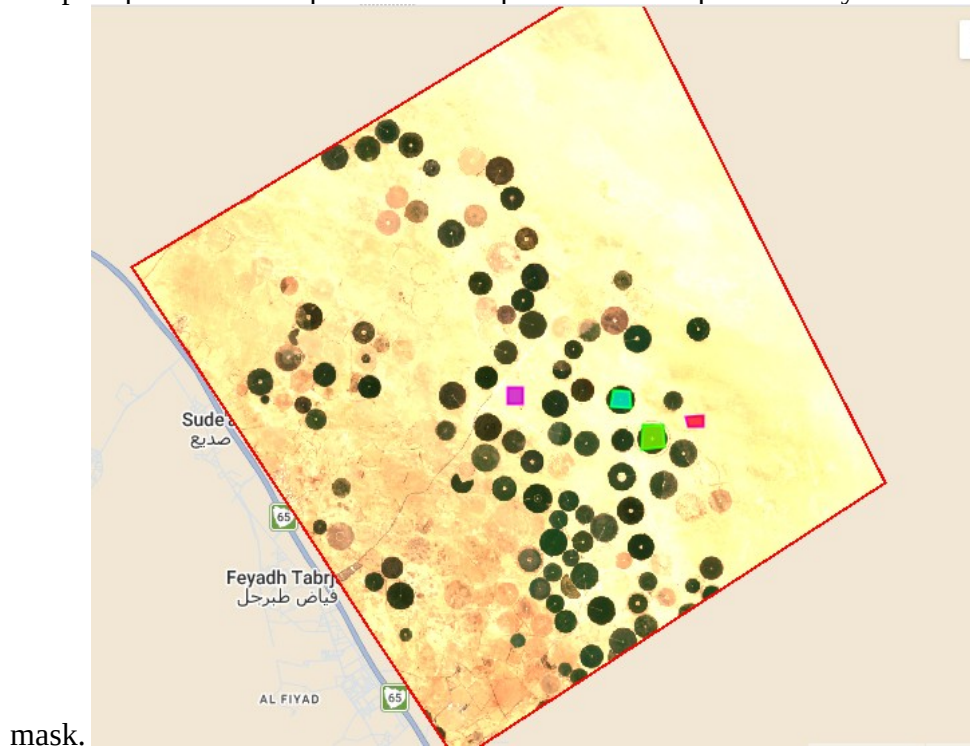


Figure 15. Manually selected reference polygons used for the light quantitative validation in April 2023. Crop reference polygons were placed inside homogeneous active center-pivot fields, whereas non-crop reference polygons were placed over homogeneous bare desert surfaces. The reference polygons were selected from the true-colour composite without using the crop-mask overlay.

Table 3. Design of the manually labelled reference samples used for the light quantitative validation.

Reference class	Number of polygons	Sampled pixels	Selection basis
Crop	2	100	Homogeneous interiors of active center-pivot fields
Non-crop	2	100	Homogeneous bare desert areas away from pivots, roads, and boundaries
Total	4	200	Manually labelled reference pixels

3.9.2 Accuracy Assessment Results

The manually labelled reference pixels were compared with the predicted classes of the NDVI-based crop mask for April 2023. The resulting confusion matrix is presented in Table 4. Out of 200 validation pixels, 198 pixels were classified correctly.

All 100 non-crop reference pixels were correctly classified as non-crop. Of the 100 crop reference pixels, 98 were correctly classified as crop, while 2 crop reference pixels were classified as non-crop. No non-crop reference pixel was incorrectly classified as crop.

The overall accuracy of the light quantitative validation was 99.0%, and the Kappa coefficient was 0.98. Producer's accuracy was 100% for non-crop and 98% for crop. User's accuracy was 98.04%

for non-crop and 100% for crop. These results indicate a strong agreement between the manually labelled reference samples and the NDVI-based crop-mask classification for the selected homogeneous validation areas.

Table 4. Confusion matrix of the April 2023 light quantitative validation. Rows represent manually labelled reference classes, and columns represent the predicted classes from the NDVI-based crop mask.

Reference class	Predicted non-crop	Predicted crop	Total
Non-crop	100	0	100
Crop	2	98	100
Total	102	98	200

Table 5. Accuracy metrics derived from the April 2023 light quantitative validation.

Accuracy metric	Non-crop	Crop	All validation samples
Producer's Accuracy	100.00%	98.00%	—
User's Accuracy	98.04%	100.00%	—
Correctly classified pixels	100 / 100	98 / 100	198 / 200
Overall Accuracy	—	—	99.00%
Kappa Coefficient	—	—	0.98

3.9.3 Interpretation and Limitations

The validation results show that the NDVI-based crop mask performed well within the selected reference areas for April 2023. The crop mask correctly identified all manually labelled non-crop pixels and correctly classified 98 out of 100 manually labelled crop pixels. This indicates that the fixed threshold of $\text{NDVI} \geq 0.40$ was effective for separating clearly active center-pivot fields from homogeneous bare desert surfaces during the peak-season period.

However, the reported overall accuracy of 99.0% should be interpreted as agreement within the selected validation samples rather than as an absolute accuracy value for the entire study area or for all months. The reference polygons were manually selected from the April 2023 true-colour composite and were located in clearly homogeneous crop and non-crop areas. Therefore, the validation samples do not fully represent mixed pixels, field boundaries, sparse vegetation, roads, or partially active fields.

In addition, the validation was performed only for one peak-season month and did not use independent ground-survey data or parcel-based reference data. The use of a fixed NDVI threshold may also lead to underestimation or overestimation in months with different crop conditions, irrigation levels, soil background effects, or residual cloud and shadow contamination. Despite these limitations, the validation results provide supporting evidence that the NDVI-based crop-mask approach is suitable for tracking broad seasonal changes in active center-pivot agriculture within the study area.

4. Conclusion

This study developed and applied an SCL-masked, rule-based NDVI crop-masking workflow to monitor active center-pivot agriculture in Wadi As-Sirhan, Saudi Arabia, between January 2020 and December 2024. Sentinel-2 Level-2A imagery was processed in Google Earth Engine using SCL-based cloud, cirrus, and shadow masking. Monthly median NDVI composites were produced for all 60 calendar months, and a fixed threshold of $\text{NDVI} \geq 0.40$ was used to generate binary crop masks representing active crop or vegetation pixels.

The results showed that Sentinel-2 data availability was sufficient throughout the full study period. A total of 2,158 Sentinel-2 scenes were available over the AOI, and no source-data gap was detected. The five-year crop-area time series revealed a clear seasonal pattern in active center-pivot agriculture. Mapped cropped area ranged from approximately 5.21 km² in January 2020 to 20.90 km² in April 2023, with relatively high values generally occurring during the spring period.

The comparison of off-season, transition-season, and peak-season months showed that the spatial distribution of NDVI-based crop pixels was visually consistent with the circular center-pivot fields visible in the true-colour imagery. In addition, the light quantitative validation for April 2023 showed 99.0% overall agreement between the crop mask and the manually labelled reference samples.

Although the results are promising, the reported accuracy represents agreement within selected homogeneous reference areas rather than an independent field-based assessment for the entire study area. Future work could improve the method by testing adaptive or season-specific NDVI thresholds, including temporal crop-profile rules, detecting pivot geometry automatically, or using independent parcel or field reference data. Nevertheless, the workflow provides a reproducible and practical approach for monitoring broad seasonal changes in active center-pivot agriculture using Sentinel-2 imagery.

5. References

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